

Test Plan – Pega Claims Application (MVP1)

1. Document Control

Field	Details
Project	Pega Claims Application – Self-Service Portal
Release	MVP1
Test Type	Manual Testing (No automation in scope)
Prepared By	[Tester/QA Lead Name]
Reviewed By	[Business/Tech Lead Name]
Version	1.0
Date	[Date]

2. Introduction

This document defines the test approach, scope, scenarios, and exit criteria for MVP1 of the Pega Claims Application. MVP1 covers anonymous self-service claim submission, automatic case creation in Pega, backend policy validation via email, and AI-based image analysis and policy data extraction.

3. Objective

To validate that:

- Users can submit claims anonymously through the self-service web form without functional errors.
 - Submitted data correctly creates a case in Pega with accurate field mapping.
 - The system correctly triggers an email to the backend team/system for policy validation.
 - AI-based analysis correctly processes uploaded images and extracts policy details with acceptable accuracy.
 - The end-to-end flow works reliably under normal and edge-case conditions.
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4. Scope

4.1 In Scope

- Self-service portal – anonymous access (no login/authentication)
- Web form – field-level validation, file/image upload, submission
- Pega case creation – triggered from form submission
- Case data mapping – form fields → Pega case fields
- Outbound email generation for policy validation (to backend team/system)
- AI image analysis (e.g., damage photos, documents)
- AI-based extraction of policy details (OCR/NLP extraction accuracy)
- Case status/tracking visibility (if available in MVP1)
- Notifications (acknowledgement email/SMS to claimant, if in scope)
- Basic security testing around anonymous access (since there's no login)

4.2 Out of Scope

- Automation testing (explicitly excluded for this phase)
 - Performance/load testing (unless separately planned)
 - Integration with downstream claims settlement systems (if not part of MVP1)
 - Multi-language support (unless confirmed in scope)
 - Regression of legacy systems not touched by this release
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5. Test Strategy

Since this is manual testing only, the strategy relies on structured test case design, exploratory testing, and strong defect tracking discipline.

5.1 Types of Testing to Perform

Test Type	Purpose
Functional Testing	Validate each feature works as per requirements (form, case creation, email, AI extraction)
Field/Data Validation Testing	Mandatory fields, formats, character limits, special characters
Negative Testing	Invalid inputs, missing files, wrong file types, malformed data
UI/UX Testing	Layout, responsiveness, error message clarity, accessibility basics

Test Type	Purpose
Cross-Browser/Device Testing	Chrome, Edge, Safari, Firefox; mobile/tablet since it's public-facing
Integration Testing	Portal → Pega case creation → email trigger → AI service handoff
AI Accuracy/Output Validation	Correctness of extracted policy data and image analysis results
Security Testing (basic)	Anonymous access abuse, file upload vulnerabilities, spam/bot submissions, injection attempts in text fields
Boundary Value Testing	Max file size, max text length, max number of attachments
Exploratory Testing	Unscripted testing to catch unknown-unknowns, especially around AI behavior
End-to-End (E2E) Testing	Full claim lifecycle from submission to case creation to backend email to extracted data landing in Pega
Regression Testing (manual)	Re-test impacted areas after each build/bug fix
Usability Testing	Since users are anonymous/public, form must be intuitive without guidance

6. Test Environment

Item	Details
Test Environment URL	[QA/UAT env link]
Pega Version	[Version]
Browsers	Chrome, Edge, Firefox, Safari (latest 2 versions)
Devices	Desktop, Android mobile, iOS mobile, Tablet
Email System	[Test mailbox/SMTP config for backend validation emails]
AI Service	[Name of AI/OCR service – internal Pega AI, or third-party API]
Test Data	Sample claim forms, sample images (clear, blurry, wrong format, tampered), sample policy documents
Access	Anonymous – no credentials, so ensure test env allows anonymous access same as prod

7. Roles & Responsibilities

Role	Responsibility
QA Tester	Test case design, execution, defect logging
Business Analyst	Requirement clarification, sign-off on expected behavior
Pega Developer	Fix defects, support test data/case configuration
AI/ML Team	Clarify expected AI extraction accuracy thresholds
Product Owner	UAT sign-off, prioritize defects

8. Detailed Test Scenarios by Module

8.1 Self-Service Portal – Anonymous Access

- Portal loads without requiring login/credentials.
- Session handling for anonymous users (timeout behavior, session ID uniqueness).
- Multiple anonymous users submitting simultaneously don't cross-contaminate data.
- Direct URL access / bookmarking behavior.
- Browser back/refresh behavior mid-form-fill (data loss handling).
- CAPTCHA or bot-prevention validation, if implemented.

8.2 Web Form – Functional & Validation

- All mandatory fields enforce validation (claimant name, policy number, incident date, etc.).
- Field-level validations: date formats, numeric-only fields, email format, phone number format.
- Character limits (min/max) on text fields.
- Dropdowns/radio buttons/checkboxes populate and behave correctly.
- File upload: supported formats (jpg, png, pdf), unsupported format rejection, file size limit enforcement.
- Multiple file uploads (if allowed) – count limits.
- Form save/draft functionality (if applicable).
- Submit button disabled until mandatory fields are filled (if that's the design).
- Error messages are clear and field-specific.
- Special character/injection handling in free-text fields (SQL injection, script tags – security angle).
- Duplicate submission handling (double-click on submit, resubmission of same claim).

8.3 Pega Case Creation

- Case gets created immediately upon form submission.
- Case ID/reference number generated and displayed/communicated to user.
- All form field values map correctly to corresponding Pega case fields (do a field-by-field mapping matrix).
- Case status is set correctly at creation (e.g., "New" / "Pending Validation").
- Attachments from the form (images/documents) are correctly attached to the case.
- Case creation failure handling – what happens if Pega is down or times out (user-facing error, retry mechanism).
- Duplicate case prevention for repeated submissions.
- Audit trail/history log entry created for case creation event.

8.4 Backend Email Trigger (Policy Validation)

- Email is triggered automatically upon case creation.
- Email is sent to the correct backend recipient/distribution list.
- Email contains correct and complete case details (policy number, claimant info, attachments/links).
- Email subject line/format follows defined template/standards.
- Failure scenario: what happens if email server is down – is there a retry or alert mechanism?
- No duplicate emails sent for a single case.
- Attachments (images/docs) are correctly embedded/linked in the email.
- Time lag between case creation and email dispatch (should meet expected SLA, even if performance testing is out of scope, basic timing check is useful).

8.5 AI Image Analysis

- Upload a clear, valid image – verify AI correctly analyzes/classifies it (e.g., damage type, image quality check).
- Upload a blurry/low-resolution image – verify system handles gracefully (rejects or flags for manual review, doesn't crash).
- Upload an irrelevant image (e.g., random photo, not related to claim) – verify AI flags mismatch.
- Upload correct format but corrupted file – error handling.
- Upload multiple images – verify all are analyzed, not just the first one.
- Verify AI output feeds correctly into the case (fields updated, or a report attached).
- Test with edge-case images: extreme lighting, multiple objects, cropped images.
- Define and validate against an "acceptable accuracy threshold" agreed with business/AI team (since 100% AI accuracy isn't realistic).

8.6 AI Extraction of Policy Details

- Upload a clear policy document – verify key fields (policy number, holder name, coverage type, dates) are extracted accurately.
- Upload a scanned/handwritten or poor-quality document – check fallback behavior (manual review flag vs. silent failure).
- Upload a document in an unexpected format/layout – verify system doesn't break, and handles gracefully.
- Cross-verify extracted data against actual document data (build a test data set with "expected extraction results" to compare against).
- Verify partial extraction handling – if only some fields are extracted, does the system flag incomplete records instead of assuming success?
- Test with documents containing multiple policies/pages – correct data isolation.
- Validate what happens when extraction confidence score is low – is there a manual review queue?

8.7 End-to-End Flow

- Full journey: User fills form → uploads image/policy doc → submits → case created in Pega → email sent to backend → AI analysis triggered → extracted data updates case → (if applicable) claimant notified.
- Validate data consistency across all these steps (no data loss/mismatch between stages).
- Time taken for complete flow (informal, since performance testing is out of scope).
- Test with multiple concurrent E2E submissions to check for cross-case data leakage.

8.8 Notifications (if in scope for MVP1)

- Acknowledgement email/SMS sent to claimant post-submission.
 - Correct case reference number communicated.
 - Notification triggers only once per submission.
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9. Test Data Requirements

- Valid and invalid sample claim data sets
 - Sample images: valid claim-relevant images, irrelevant images, blurry/corrupted images, oversized images
 - Sample policy documents: clear scans, poor-quality scans, wrong document type
 - Boundary test data: max character limits, max file sizes, special characters
 - A "golden dataset" with pre-verified expected extraction results, to measure AI accuracy objectively
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10. Entry Criteria

- Build deployed to QA/UAT environment and stable
- Test cases reviewed and approved
- Test data prepared
- Access to Pega case view, email test mailbox, and AI service logs available for verification

11. Exit Criteria

- All planned test cases executed
- No open Critical/High severity defects
- Agreed AI accuracy threshold met (e.g., $\geq X\%$ correct extraction, as agreed with business)
- Defect resolution rate meets agreed threshold (e.g., 95% of Medium/Low defects fixed or deferred with sign-off)
- UAT sign-off obtained from Product Owner/Business

12. Defect Management

- Tool: [JIRA / Azure DevOps / Pega ALM, etc.]
- Severity classification: Critical / High / Medium / Low
- Priority classification: P1-P4
- Defect lifecycle: New → Assigned → Fixed → Retest → Closed/Reopened
- Daily defect triage during test cycle

13. Risks & Mitigations

Risk	Mitigation
AI extraction accuracy not meeting expectations	Define acceptable threshold upfront with business; build manual review fallback
No automation – regression takes longer with each build	Prioritize regression suite to critical paths only
Anonymous access increases spam/bogus submissions	Ensure basic bot-protection (CAPTCHA/rate limiting) testing is included
Email delivery failure to backend team	Test failure/retry scenarios explicitly, not just happy path
Ambiguous field-mapping between form and Pega case	Get a field-mapping document/matrix from BA before testing starts

Risk	Mitigation
Image/document quality varies widely in real world	Build a diverse, realistic test data set upfront

14. Assumptions & Dependencies

- Field-to-case mapping document will be provided by the Pega development team.
- Expected AI accuracy threshold will be defined and agreed upon by business/AI team before test sign-off.
- Test environment will have anonymous access enabled, mirroring production.
- Backend email recipient/test mailbox will be accessible to QA for verification.
- Sample/golden test data for AI validation will be prepared jointly with AI/ML team.

15. Test Deliverables

- Test Plan (this document)
- Test Case document (detailed step-by-step, with expected results)
- Traceability Matrix (requirement ↔ test case mapping)
- Defect Report
- Test Summary Report (post-execution)
- AI Accuracy Validation Report (extraction/analysis results vs. golden dataset)

16. Sign-off

Name	Role	Signature	Date
	QA Lead		
	Product Owner		
	Business Analyst		